

Propulsion Architecture Trade Study for a Small VTOL ISR UAV

Project Outline (1–2 Week Engineering Study)

1. Project Goal

Perform a system-level propulsion architecture trade study for a small VTOL fixed-wing ISR UAV. The objective is to compare competing VTOL propulsion concepts using physics-based reasoning, mass and power estimates, and qualitative system complexity assessment. No hardware is built.

~~2. Final Deliverable~~

~~One professional PDF report (8–12 pages) suitable for an internship application portfolio. The report documents assumptions, calculations, trade-offs, and a final engineering recommendation.~~

3. Mission Definition

Define a realistic but fictional UAV mission that anchors all calculations. Typical parameters include MTOW, payload mass, endurance, cruise speed, operating altitude, and VTOL takeoff/landing requirement.

4. Architectures Compared

Compare at least two VTOL propulsion architectures:

- Quad-lift + pusher prop (baseline)
- Tilt-rotor configuration

~~An optional third concept (e.g. tail-sitter) may be included if time permits.~~

5. Block Diagrams

Create one block diagram per architecture showing major subsystems and their interactions. Blocks include energy storage, power distribution, lift propulsion, cruise propulsion, flight control, sensors, and payload. Diagrams are high-level and non-geometric.

6. Hover Analysis

Estimate hover thrust and power requirements for each architecture. Compute thrust per motor, disk loading, and hover power using simplified actuator disk theory. Clearly state all assumptions.

7. Cruise Analysis

Estimate cruise performance assuming steady level flight. Compute wing loading, lift requirement, drag estimate, and cruise shaft power. Discuss the impact of inactive lift motors during cruise where applicable.

~~8. Mass and Complexity Assessment~~

~~Estimate mass penalties for motors, ESCs, structural reinforcements, and mechanical systems. Discuss qualitative factors such as mechanical complexity, failure modes, redundancy, and~~

maintenance implications.

9. Trade-Off Matrix

Summarize results in a trade-off table comparing hover power, cruise efficiency, mass efficiency, mechanical complexity, and failure tolerance. Justify each rating with quantitative or qualitative arguments.

10. Engineering Conclusion

Provide a clear recommendation for the preferred architecture given the defined mission. Explain why it is favored and under what conditions another architecture might become preferable.

~~11. Optional Extensions~~

~~If time allows, include sensitivity studies such as MTOW variation, hot day hover performance, or battery mass fraction effects. These are optional but demonstrate deeper insight.~~

~~12. Suggested Timeline~~

~~Day 1: Mission definition and architectures~~

~~Day 2: Block diagrams~~

~~Day 3: Hover analysis~~

~~Day 4: Cruise analysis~~

~~Day 5: Mass and complexity assessment~~

~~Day 6: Trade-off matrix and conclusions~~

~~Day 7: Final report formatting~~